33 kV FROM HV SWITCHBOARD
F00-SW-010
2000-F00-SLD-EL-10004

F28-SR-002
415V ULTRA FINE GRINDING SWITCHROOM

F28-TR-009
TRANSFORMER ROOM

F28-TR-009
33/0.433kV
1000kVA
Dyn11
z=6%
ANAF

NOTE 3

SWR EARTH BAR
Tx EARTH BAR
MCC EARTH BAR
NEUTRAL BAR

1600/ 1A
2.5VA
5P20

F28-SW-009
ULTRA FINE GRINDING 415V MOTOR CONTROL CENTRE

NOTE 1
1/700A
1600/ 5A
30VA
5P20

00F-01
1600/ 5A
30VA
C10.5
TEST
PM
20A
63A
2A
32A
00F-08
ACB
1600A
AFD
240VAC SPRING
CHARGE SUPPLY
24VDC CONTROL POWER SUPPLY
NOTE 7
100A
100A
SD
NOTE 6

415V 3NE~ 50Hz 1600A 65kA for 1sec

2F-09	3F-11	4F-12	2R-09	3R-13	3R-17	INC	1F-13	2F-13	4F-16	2R-13	3R-21
20A	20A	20A	20A	20A	20A		400A	20A	20A	20A	20A
2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A		2A 32A	2A 32A	2A 32A	2A 32A	2A 32A
70A	70A	70A	70A	70A	70A		400A	70A	70A	70A	70A
EOL	EOL	EOL	EOL	EOL	EOL		EOL	EOL	EOL	EOL	EOL

CONTINUED ON SHT 2
2000-F28-SLD-EL-10202

FEEDER TO F28-SW-109
2000-F28-SCM-EL-10208

E-SHUT
F28SR001-LCS
NOTE 12

F28-FN-481	F28-FN-482	F28-FN-483	F28-FN-484	F28-FN-485	F28-FN-486	F28-HT-481	F28-HT-482	F28-HT-484	F28-HT-485
0.55kW	2.2kW	2.2kW	0.55kW	2.2kW	2.2kW	2.4kW	3.2kW	2.4kW	3.2kW
UFG MILL 2 BEARING OIL COOLER FAN 1	UFG MILL 2 GEARBOX OIL COOLER FAN 1	UFG MILL 2 GEARBOX OIL COOLER FAN 2	UFG MILL 3 BEARING OIL COOLER FAN 1	UFG MILL 3 GEARBOX OIL COOLER FAN 1	UFG MILL 3 GEARBOX OIL COOLER FAN 2	UFG MILL 2 BEARING OLS HEATER	UFG MILL 2 GEARBOX OLS HEATER	UFG MILL 3 BEARING OLS HEATER	UFG MILL 3 GEARBOX OLS HEATER
2000-F28-SCM-EL-10210	2000-F28-SCM-EL-10211	2000-F28-SCM-EL-10212	2000-F28-SCM-EL-10213	2000-F28-SCM-EL-10214	2000-F28-SCM-EL-10215	2000-F28-SCM-EL-10216	2000-F28-SCM-EL-10217	2000-F28-SCM-EL-10218	2000-F28-SCM-EL-10219

NOTES

1. EARTH-NEUTRAL LINK SHALL BE INSTALLED IN INCOMER TIER. SIGNAGE SHALL BE PLACED ON MCC AND ON TRANSFORMER READING "M.E.N. LINK LOCATED IN F28-SW-009 INCOMER TIER" CABLE SHALL BE IDENTIFIED C/W LABEL STATING "M.E.N. LINK - DO NOT REMOVE".
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. UPSTREAM TRANSFORMER FEEDER SHALL BE TRIPPED ON ACTIVATION OF SWITCHROOM EMERGENCY PUSHBUTTON, DETECTION OF EARTH FAULT OR ARC FAULT DETECTION.
4. SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
5. ALL DRIVE I/O VIA COMMS NETWORK TO DCS. FOR COMMUNICATIONS BLOCK DIAGRAM REFER 2000-F28-BLK-IC-10001.
6. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
7. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
8. THERMAL OVERLOAD UNITS MAY BE SUBSTITUTED WITH ELECTRONIC (MAGNETIC) OVERLOAD UNITS TO SUIT DUTY.
9. TOLs TO BE MOUNTED IN PURPOSE-BUILT ENCLOSURE BENEATH VSD. TOLs MAY BE INCLUDED IN MCC MODULE WITH ENGINEERING APPROVAL.
10. ANY DRIVE RTDs SHALL BE WIRED TO DEDICATED REMOTE I/O CARDS IN THE FIELD. RTD RELAYS MAY BE INCLUDED IN THE MCC MODULE WITH ENGINEERING APPROVAL.
11. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
12. EMERGENCY SHUTDOWN (E-SHUT) PUSH BUTTON N/O WIRED TO SHUNT TRIP NOMINATED CIRCUIT BREAKERS.

LEGEND

- DIRECT ON-LINE MOTOR STARTER GENERIC - NOTE 4, 5
- DIRECT ON-LINE MOTOR STARTER REVERSING - NOTE 4, 5
- EOL — ELECTRONIC OVERLOAD UNIT NOTE 4, 5
- THERMAL OVERLOAD UNIT NOTE 8
- VARIABLE SPEED DRIVE NOTE 4, 5
- I > — EARTH LEAKAGE RELAY NOTE 4, 5
- FIELD-MOUNTED DECONTACTOR
- MOTOR THERMISTOR (ONLY REQ. FOR MOTORS >37kW - 250kW)
- RTD — MOTOR RTD (ONLY REQ. FOR MOTORS >250kW)
- MOTOR HEATERS (ONLY REQ. FOR MOTORS >90kW)
- 49 63 71 — TRANSFORMER PROTECTIVE DEVICES (REFER ANSI STANDARDS FOR DESCRIPTIONS)
- CIRCUIT BREAKER (NOTE 11)
- xxxA — ST TRIP SET POINT
- xxxA — CB SIZE
- AFD — ARC FLASH DETECTION
- ECR — EARTH CONTINUITY RELAY
- E-SHUT — SWITCHROOM EMERGENCY SHUTDOWN
- 63A 3 PHASE SOCKET OUTLET

2000-F28-SLD-EL-10201_Rev2

Ultra fine grinding 2 / 3 F28-SW-009 415V MCC single line diagram sheet 1

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\ELEC\F28 - ULTRAFINE GRINDING 2 & 3\2000-F28-SLD-EL-10201.DWG 29SEP26 7:30AM ROBIN.LAW